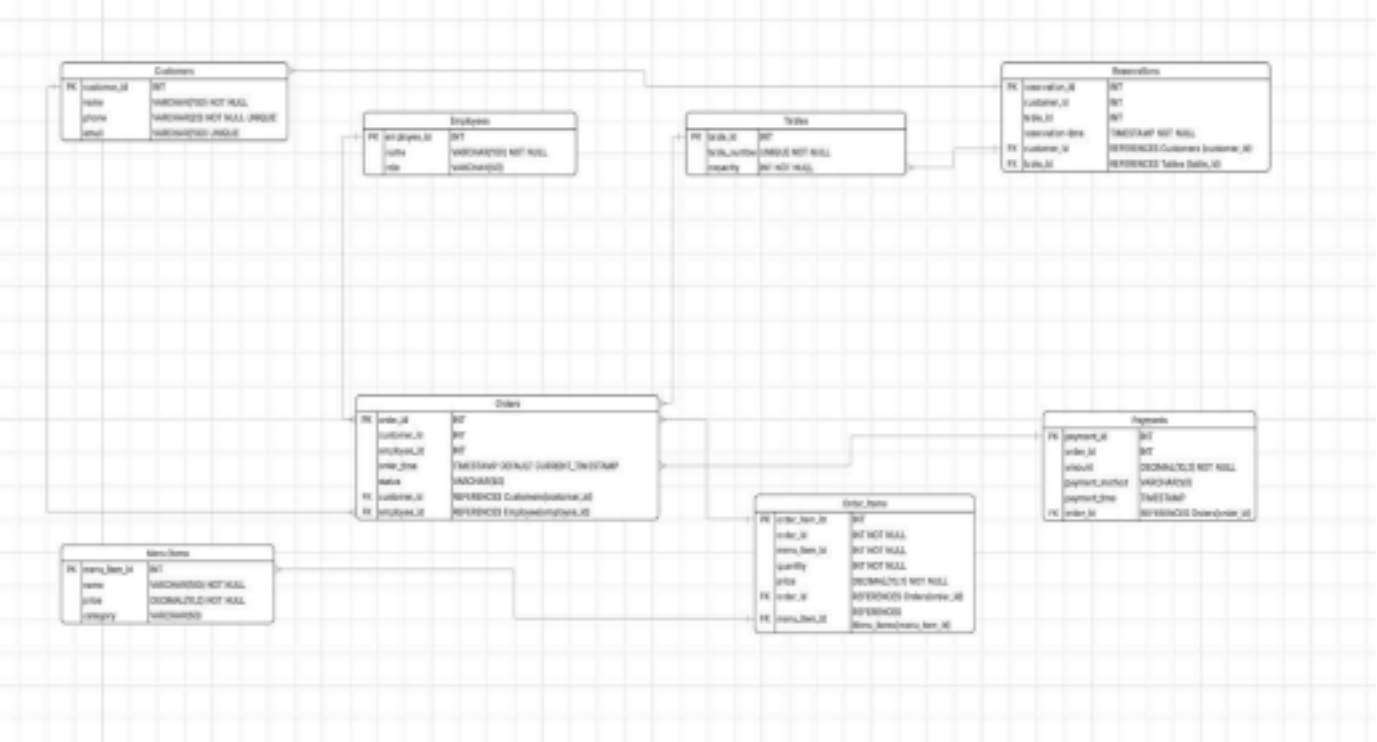
1.1 Logic



al Scheme

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1.2 Objects

Table Description

Stores detailed information about restaurant customers including contact details, used for managing reservations and orders.

Customers	customer_id	Unique ID	Int PK

	name	Customer name	varchar
	Phone	phone number	varchar
	email	Email	varchar

Comments on table relationships

A one-to-many relationship exists between Customers and Orders, where a single customer can place multiple orders

Example with data

1	Ayat	77788718721	ayat@mail.com
2	Temirlan	88775523432	temirlan@mail.com
3	gizatov	77765645554	gizatov@mail.com

4 sanzhik 77788718767 sanzhik@mail.com

Stores information about restaurant employees including roles and employment details.

Empoloyees	employee_id	Unique,pk	INT, PK, SERIAL

	name	employee name	VARCHAR(100), NOT NULL

role Job role (waiter

manager chef etc)

3 gizatov chef

4 sanzhik waiter

1	Ayat
2	Temirlan

One employee can handle many orders.

Stores restaurant tables and their seating capacity.

VARCHAR(50)

One table can be used in many reservations.

Tables	table_id	Unique,pk	Int
	table_number	Table number	UNIQUE, NOT NULL
	capacity	Number of seats	int not null

1	1	4
2	2	5

3 3 6

4 4 3

5	5	8

Reservations establish many-to-one relationships with both Customers and Tables. Each reservation is made by one customer and assigned to one table

reservations	reservation_id	Unique,pk	Int,pk,SERIAL
	customer_id	customer reference	INT, FK
	table_id	Reserved table	int not null
	reservation time	date and time ofreservation	TIMESTAMP, NOT NULL

1	1	1	2026-04-08 18:00
2	2	3	2026-04-08 19:00

3 3 5 2026-04-08 20:00 4 4 2 2026-04-09 18:30

5	5	1	2026-04-09 19:30

Menu_Items are connected to Order_Items through a one-to-many relationship, where each menu item can be included in multiple orderitems

menu items	menu_item_id	Unique,pk	int
	name	Name of the dish	VARCHAR(100), NOT NULL
	price	Price of item	DECIMAL(10,2), NOT NULL
	category	food category	VARCHAR(50)

1	Burger	1500	Main Dish
2	Cola	500	Drink

3	Pizza	3500	Main Dish
4	Water	300	Drink
5	shawerma	1700	Side Dish

Orders maintain many-to-one relationships with both Customers and Employees single customer and handled by one employee, both customers and employees can be associated with multiple order

orders	order_id	Unique,pk	int
	customer_id	Customer who made order	INT, FK

	employee_id	Employee handling order	INT, FK
	order_time	Time of order	TIMESTAMP
	status	Order status	VARCHAR(50)

Many orders belong to one customer.

Many orders are handled by one employee.

1	1	1500	2026-04-08 18:05	Completed
2	2	500	2026-04-08 19:10	Pending
3	3	3500	2026-04-08 20:20	Completed
4	4	300	2026-04-09 18:45	Completed
5	5	1700	2026-04-09 19:30	Pending

Order_Items act as an associative entity resolving a many-to-many relationship between Orders and Menu_Items (Each record links one order)

orders	order_item_id	Unique,pk	int

	order_id	Related order	INT not null
	menu_item_id	Ordered item	INT not null
	quantity	Quantity	int not null
	price	Price per item	DECIMAL(10,2)

1	1	1	2	2500
2	2	2	3	3000

3	3	3	1	1500

4 4 4 2 300

5 5 5 4 667

Stores payment information for customer orders.

a one-to-one relationship, where each payment corresponds to a specific order

Payments	payment_id	Unique,pk	int
	order_id	Related order	INT
	amount	Ordered item	DECIMAL(10,2), NOT NULL
	payment_method	Quantity	VARCHAR(50)
	payment_type	Price per item	TIMESTAMP

1	1	5500	Card	2026-04-08 18:05
2	2	4500	Card	2026-04-08 19:10
3	3	3000	Cash	2026-04-08 20:20
4	4	900	Cash	2026-04-09 18:45
5	5	1800	Card	2026-04-09 19:30